

session6_hw_tsa

```
library(fpp3)
```

— Attaching packages

fpp3 1.0.3 —

✓ tibble	3.3.1	✓ tsibble	1.2.0
✓ dplyr	1.2.1	✓ tsibbledata	0.4.1
✓ tidyr	1.3.2	✓ ggtime	1.0.0
✓ lubridate	1.9.5	✓ feasts	0.5.0
✓ ggplot2	4.0.3	✓ fable	0.5.0

— Conflicts —

fpp3_conflicts —

* lubridate::date()	masks	base::date()
* dplyr::filter()	masks	stats::filter()
* tsibble::intersect()	masks	base::intersect()
* tsibble::interval()	masks	lubridate::interval()
* dplyr::lag()	masks	stats::lag()
* tsibble::setdiff()	masks	base::setdiff()
* tsibble::union()	masks	base::union()

Exercise 1

Using `us_retail_employment`, compute the 2×12 moving average of Employed. The data is monthly, so the seasonal period is $m = 12$, which is even. Work out `.before` and `.after` for each of the two imbalanced windows before you type anything. Plot the result over the original series.

```
# Filter the series and select relevant columns
us_retail_employment <-
  us_employment |>
  filter(year(Month) >= 1990, Title == "Retail Trade") |>
  select(-Series_ID)

us_retail_ma <- us_retail_employment |>
  mutate(
    MA_12_R = slider::slide_dbl(Employed, mean,
                                .before = 5, .after = 6, .complete = TRUE),
    MA_12_L = slider::slide_dbl(Employed, mean,
                                .before = 6, .after = 5, .complete = TRUE),
    MA_2x12 = 0.5 * (MA_12_R + MA_12_L)
  )
```

```
us_retail_ma |>
  select(Month, Employed, MA_12_R, MA_12_L, MA_2x12) |>
  head(10)
```

```
# A tsibble: 10 x 5 [1M]
  Month Employed MA_12_R MA_12_L MA_2x12
  <mth>    <dbl>    <dbl>    <dbl>    <dbl>
1 1990 Jan   13256.      NA      NA      NA
2 1990 Feb   12966.      NA      NA      NA
3 1990 Mar   12938.      NA      NA      NA
4 1990 Apr   13012.      NA      NA      NA
5 1990 May   13108.      NA      NA      NA
6 1990 Jun   13183.  13186.      NA      NA
7 1990 Jul   13170.  13170.  13186.  13178.
8 1990 Aug   13160.  13151.  13170.  13161.
9 1990 Sep   13113.  13130.  13151.  13141.
10 1990 Oct   13185.  13103.  13130.  13117.
```

```
us_retail_ma |>
  select(Month, Employed, MA_12_R, MA_12_L, MA_2x12) |>
  tail(10)
```

```
# A tsibble: 10 x 5 [1M]
  Month Employed MA_12_R MA_12_L MA_2x12
  <mth>    <dbl>    <dbl>    <dbl>    <dbl>
1 2018 Dec   16309.  15813.  15819.  15816.
2 2019 Jan   15754.  15807.  15813.  15810.
3 2019 Feb   15567.  15800.  15807.  15804.
4 2019 Mar   15577.  15794.  15800.  15797.
5 2019 Apr   15625.      NA  15794.      NA
6 2019 May   15692.      NA      NA      NA
7 2019 Jun   15776.      NA      NA      NA
8 2019 Jul   15786.      NA      NA      NA
9 2019 Aug   15750.      NA      NA      NA
10 2019 Sep   15611.      NA      NA      NA
```

At the beginning: - MA_12_R needs 5 months before + current + 6 after, so it becomes available after 5 missing rows. - MA_12_L needs 6 months before + current + 5 after, so it becomes available after 6 missing rows. - Since MA_2x12 needs both, it starts only when both exist. At the end: - One of them runs out 5 months before the end. - The other runs out 6 months before the end. - So the combined MA_2x12 also has missing values near the end.

So the final MA_2x12 loses first 6 and last 6 rows.

```
us_retail_ma |>
  autoplot(Employed, colour = "grey") +
  geom_line(aes(y = MA_2x12), colour = "brown") +
  labs(
    title = "US retail employment with 2x12 moving average",
    y = "People employed"
  )
)
```

Warning: Removed 12 rows containing missing values or values outside the scale range (``geom_line()``).

